

Cost-Benefit Analysis of Solar Adoption in Residential Households of the

MMR

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Table of Contents

Table of Contents	2
Chapter 1: Introduction	3
1.1 Background of the Study	4
1.2 Objectives of the Study	4
1.3 Scope and Limitations of the Study	4
1.3.1 Scope of the Study	4
1.3.2 Limitation of the Study	5
Chapter 2: Literature Review	6
2.1 Power on the Roof: Reviewing Rooftop Solar Adoption in India	6
2.2 Solar Rooftop Energy Installations: Cost and Benefit Analysis	6
2.3 A Study of Pradhan Mantri Surya Ghar Yojna and Its Impact on Household Electricity Costs and Decreasing Dependency on Conventional Energy Sources	7
2.4 WHO SAYS YES? Simulating How Rooftop Solar Decisions Are Taken in an Indian Housing Society	7
2.5 SOLAR ROOFTOP: PERSPECTIVE OF DISCOMS	8
Chapter 3: Methodology	9
3.1 Research Design	9
3.2 Sampling and Data Collection	9
3.3 Data Analysis Tools	10
Chapter 4: Field Work Observations and Analysis	11
4.1 Demographic and Housing Profile and Adoption Status	11
4.2 Analysis of Barriers for Non-Adopters	12
4.2.1 The Upfront Cost	13
4.2.2 Information and Awareness Deficits	13
4.2.3 Structural and Spatial Limitations	14
4.3 Cost-Benefit Analysis	16
4.3.1 Capital Expenditure and Subsidy	16
4.3.2 Operational Savings, Discounted Payback Period, and NPV	16
Chapter 5: Conclusion and Recommendations	18
5.1 Conclusion	18
5.2 Recommendations	19
5.2.1 For Policy Makers	19
5.2.2 For Distribution Companies (DISCOMs)	19
5.2.3 For Solar Vendors	19
5.2.4 For Cooperative Housing Societies	20
Bibliography	21
Appendix	22
Appendix A: Field Survey Questionnaire	22
Appendix B: 25-Year Discounted Cash Flow (DCF) Model	24

Chapter 1: Introduction

The Pradhan Mantri Surya Ghar: Muft Bigli Yojana (PMSG: MBY) launched on February 13, 2024, represents a forward step in India's direction towards renewable energy generation and net-zero emissions. The primary objective of the program is to provide one crore households across the country with three hundred units of free electricity every month through the installation of grid rooftop solar (RTS) systems.

To make rooftop solar more accessible and affordable, the scheme has a financial outlay of ₹75,021 crore. The financial framework of PMSGY is based on the subsidy or Central Financial Assistance (CFA) provided directly into the consumer's account. The subsidy is tiered based on the capacity installed capping at 3kW:

1. For systems up to 2 kW: A subsidy of ₹30,000 per kW is provided.
2. For systems between 2 kW and 3 kW: An additional subsidy of ₹18,000 per kW is provided for the incremental capacity.
3. Maximum Cap: The total subsidy for systems larger than 3 kW is strictly capped at ₹78,000.
4. Group Housing Societies/Resident Welfare Associations (GHS/RWA): Subsidies are also extended to common facilities, including EV charging, at ₹18,000 per kW up to a 500 kW capacity (calculated at 3 kW per house)

The scheme also includes provisions for collateral-free, low interest loans at the rate of around seven per cent.

A core feature of PM Surya Ghar Yojana is the fully online, transparent and demand-driven implementation process. It is managed via the National Portal (pmsuryaghar.gov.in) and the PM-Surya Ghar mobile application which offers features such as:

1. Estimation tools for suitable RTS system sizes and financial parameters.
2. A directory to view and select registered installation vendors in the area.
3. Direct API integration with local Distribution Utilities (DISCOMs) to fetch consumer data, process technical feasibility, and facilitate net-metering agreements.
4. A streamlined regulatory process that waives the technical feasibility requirement for certain installations and introduces auto-load enhancement up to 10 kW

Upon successful implementation and inception by the DISCOM, the CFA is released into the applicant's bank account within 15 to 30 days.

As of March 19, 2026, the scheme has successfully installed 26,19,879 RTS with ₹17,967.53 crore disbursed as Central Financial Assistance. The scheme is projected to save the government an estimated ₹75,000 crores annually in power procurement and subsidy costs. It also creates vast opportunities for local manufacturers, skilled workers, and various industries engaged in the solar energy value chain.

1.1 Background of the Study

India has seen a rise in the demand for renewable energy to reduce fossil fuel dependency. To address this the Indian government established ambitious targets at 2021 United Nations Climate Change Conference (COP26) under the “Panchamrit” framework to achieve 500 GW of non-fossil fuel based electricity by 2030 and reach Net Zero emissions by 2070.

The solar project has seen an exponential growth but it was predominantly led by the Commercial and Industrial (C&I) sector and the decentralised rooftop solar segment fell short of the national targets. To rectify this, the Government of India launched the Pradhan Mantri Surya Ghar: Muft Bijli Yojana in February 2024.

In many cities, RTS adoption is highly dependent on collective decisions within housing societies or Resident Welfare Associations (RWAs). In the context of the Mumbai Metropolitan Region (MMR) and Maharashtra, local distribution utilities like the BEST Undertaking and Mahavitaran (MSEDCL) are actively facilitating the scheme's rollout. Research indicates that many households perceive rooftop solar primarily through the lens of financial risk and uncertainty regarding long-term returns, rather than purely as an environmental action.

This study aims to conduct a Cost-Benefit Analysis of solar energy adoption under the PM Surya Ghar Yojana in the MMR, to understand the real-world viability and challenges of residential solar transitions.

1.2 Objectives of the Study

The specific objectives of this study are structured as follows:

1. To assess current adoption rates specifically in Cooperative Housing Societies (CHS) and Independent Houses/Bungalows.
2. To identify barriers to adoption
3. To measure awareness of government initiatives
4. To evaluate financial viability and consumer readiness
5. To provide actionable insights

1.3 Scope and Limitations of the Study

1.3.1 Scope of the Study

The scope of this field project is defined by its geographical, demographic, and thematic boundaries:

1. Geographical Scope: The research is strictly confined to the Mumbai Metropolitan Region (MMR).
2. Demographic Scope: Consumers of two primary residential profiles: those living in Flats/Apartments within Cooperative Housing Societies (CHS) and those residing in Independent Houses, Bungalows, or Row Houses.
3. Thematic Scope: It analyzes average monthly electricity bills, upfront installation costs, and the financial viability of solar systems given the current Central Financial Assistance (CFA) available under the PM Surya Ghar: Muft Bijli Yojana. It also investigates the awareness of government subsidies, and the willingness of residents to participate in shared community solar initiatives.
4. Methodological Scope: The study relies on primary data collected through a structured field survey administered to MMR residents.

1.3.2 Limitation of the Study

While the study aims to provide a comprehensive cost-benefit analysis, certain limitations must be acknowledged:

1. **Sample Size and Selection Bias:** As the primary data is collected via a convenience survey circulated among residents, the sample may be subject to self-selection bias. The findings may represent the views of an English-speaking literate demographic and may not be perfectly representative of the entire socio-economic spectrum across the MMR.
2. **Self-Reporting Inaccuracies:** The survey relies on self-reported data regarding average monthly electricity bills and current solar adoption status. Respondents might estimate rather than provide exact figures, potentially introducing a margin of error into the financial viability calculations.
3. **Complexity of CHS Dynamics:** In Mumbai, a vast majority of the population resides in Cooperative Housing Societies. The study captures the individual resident's willingness to adopt solar energy or sign petitions for shared systems, but actual adoption in a CHS requires collective consensus and committee approvals. The survey data alone cannot account for the complex, real-world political and social negotiations within these societies.
4. **Technical and Structural Generalizations:** The study addresses some perceived barriers among non-adopters. However, without technical assessments by registered DISCOM vendors, the actual capacity for these specific households remains unknown.
5. **Policy Environment:** The regulatory framework, including state-level net-metering policies and the PM Surya Ghar: Muft Bijli Yojana subsidy structures, is dynamic. The findings of this report represent a snapshot based on the current financial parameters and awareness levels as of early-to-mid 2026.

Chapter 2: Literature Review

This chapter reviews the existing literature surrounding the financial, structural, behavioural and regulatory dimensions of rooftop solar adoption. By examining these research papers and reports, this review provides a theoretical and empirical foundation for understanding the complexities of solar energy transitions in residential settings, such as those found in the Mumbai Metropolitan Region (MMR).

2.1 Power on the Roof: Reviewing Rooftop Solar Adoption in India

This review by Saithu Mohammed and P. Maheswari conducted in 2025 represents the trajectory of rooftop solar in India. The authors highlight that while rooftop solar minimizes transmission losses and provides resilience benefits, adoption has still lagged behind national targets. Prior to recent initiatives, the commercial and industrial (C&I) sectors dominated RTS adoption due to favorable tariffs and easier access to institutional credit, while residential uptake still remained marginal. The paper points out that residential adoption only witnessed acceleration following the introduction of the PM Surya Ghar scheme, which streamlined digital pipelines and provided simplified subsidy structures. However, barriers remain primarily characterized by high upfront capital costs, affordability gaps, and regulatory uncertainty.

To address the financial capabilities and built environments across Indian cities, authors detailed three business models according to the capital costs, operational risk, and financial benefits.

1. Consumer-Owned Capital Expenditure (CAPEX) Model is a where consumer bears the upfront capital cost using savings/bank loans/government-subsidies and retains the full ownership of the system reaping lifetime financial benefits primary through offsetting high grid electricity tariffs and earning export credits via net-metering. However it is subject to the payback period.
2. Third-Party Owned Operating Expenditure (OPEX) Model removes the financial burden from the consumer as a third-party developer installs, owns and maintains the rooftop system on the consumer's residence. The consumer then purchases the generated electricity from the developer through a long-term Power Purchase Agreement (PPA) or a leasing arrangement, typically at a rate lower than the standard tariff. This model shifts the capital expenditure and maintenance to the developer and is used heavily in the Commercial and Industrial (C&I) sectors where energy requirements are high and predictable.
3. Collective and Virtual Deployment Models which includes Group Net Metering where the electricity generated from a single shared rooftop system is credited across multiple individual consumer accounts within a society and Virtual Net Metering where consumers subscribe to or purchase a share of an off-site solar installation and receive the financial credits on their personal electricity bills.

2.2 Solar Rooftop Energy Installations: Cost and Benefit Analysis

Dr. Harish Sharma's 2024 analysis explores the technological and economic variables influencing rooftop solar viability. The paper outlines the core components required for a functional system, emphasizing that beyond the solar photovoltaic (PV) panels, inverters and

net metering systems are crucial for making the generated electricity compatible with the grid.

Financially, the study acknowledges that initial installation costs represent the most significant barrier to adoption. While these upfront expenses are heavily mitigated by government subsidies and solar loans, the typical payback period for a residential installation still ranges from 5 to 10 years, depending on system size, local electricity prices, and consumption patterns. Beyond the direct savings of reduced or eliminated electricity bills, the study notes that homes with solar installations frequently experience an increase in property value. Environmentally, rooftop solar provides substantial benefits by displacing fossil fuel generation, thereby lowering greenhouse gas emissions and decreasing air pollution.

2.3 A Study of Pradhan Mantri Surya Ghar Yojna and Its Impact on Household Electricity Costs and Decreasing Dependency on Conventional Energy Sources

Neev Shah's 2024 paper evaluates the socioeconomic and environmental impacts of the PM Surya Ghar: Muft Bijli Yojana. The scheme is framed as a transformative initiative that fundamentally lowers household electricity costs while driving India towards its carbon emission reduction targets. Beneficiaries can generate and consume up to 300 units of free electricity every month, generating estimated annual savings of ₹15,000 to ₹18,000 on their bills. The research details the tiered financial support driving this transition, noting a 60% subsidy for systems up to 2kW, and a 40% subsidy for incremental capacity between 2kW and 3kW. However, the paper identifies several critical hurdles slowing widespread adoption. Household reluctance driven by the pre-existing availability of free electricity in certain states is one of them. Furthermore, physical barriers such as restricted terrace space, uneven terrain, and shading complicate installations. The study also highlights the operational strain the scheme places on Distribution Utilities (DISCOMs), who must act as unpaid storage facilities under the current net metering framework absorbing excess solar energy during the day while consumers draw from the grid at night.

2.4 WHO SAYS YES? Simulating How Rooftop Solar Decisions Are Taken in an Indian Housing Society

This 2026 gaming workshop report by Transitions Research delves into the psychological and social dynamics which govern solar adoption in collective living spaces. The report underlines that technical feasibility and financial incentives alone do not guarantee RTS adoption in Indian cities but social acceptance and collective decision-making within housing societies are equally critical.

The workshop reveals that financial risk heavily dominates early conversations among residents, with many viewing the technology purely as a financial investment rather than an environmental imperative. A critical finding of the study is the trust deficit: residents typically trust local community members and neighbors far more than they trust institutional claims or government programs. The report concludes that to break decision paralysis, stakeholders must create visible social signals, transparent lifecycle data, and frame rooftop solar as a collective community benefit.

2.5 SOLAR ROOFTOP: PERSPECTIVE OF DISCOMS

Conducted by The Energy and Resources Institute (TERI) in 2019, the report reveals that many DISCOMs perceive rooftop solar primarily as a threat to their revenue rather than an opportunity. Since the Indian electricity market is designed around a "cross-subsidy" model, high-consuming residential users, as well as commercial and industrial clients, are charged premium rates. The profits generated from these high-paying consumers are used to subsidize the cost of electricity for low-income households and the agricultural sector, who pay significantly below the actual cost of supply. Because rooftop solar requires an upfront investment and ample roof space, early adopters are almost exclusively these high-paying consumers, when they transition to solar, they drastically reduce their grid electricity purchases and the DISCOM suffers a massive reduction in its most profitable energy sales. Meanwhile, the low-paying consumers who lack the roof space or capital for solar remain entirely dependent on the grid, thereby increasing the cross-subsidy burden on the remaining non-solar consumers.

While the DISCOM's revenue drops, its operating costs do not drop proportionally. DISCOMs procure the vast majority of their electricity from conventional generating stations through long-term Power Purchase Agreements (PPAs) that often span 25 years. The billing structure of a PPA is split into two parts:

- a. Variable Charges: The cost of the actual fuel used to generate the power.
- b. Fixed Charges: The cost covering the power plant's infrastructure, debt, and maintenance.

When a household generates its own solar power, the DISCOM draws less power from the conventional grid. They save on the variable charges because less coal is burned. However, the DISCOM is required to pay the fixed charges to the power generators regardless of whether they actually use the electricity. Furthermore, state generating companies are generally unwilling to reduce the quantum of power procured by the DISCOMs. To compound the financial strain, if a solar household generates more power than it consumes, the DISCOM is legally obligated under net-metering regulations to absorb that excess power into the grid and financially compensate the consumer for it. Until new utility-centric business models are adopted or regulatory mechanisms like gross-metering are widely implemented to protect utility revenues, DISCOMs will continue to bear the financial brunt of the decentralized solar transition.

Chapter 3: Methodology

3.1 Research Design

This study employs a descriptive and analytical research design to evaluate the cost-benefit dynamics and behavioral barriers associated with rooftop solar adoption in the Mumbai Metropolitan Region (MMR). A mixed-methods approach is utilized relying on quantitative primary data to assess financial metrics and policy awareness, complemented by qualitative insights into consumer behavior and decision-making constraints.

3.2 Sampling and Data Collection

The study relies on field data collection conducted among residential electricity consumers across the MMR. The primary data collection instrument is a structured, self-administered questionnaire. Prior to answering the core questions, all respondents were required to provide informed consent, ensuring the ethical collection of academic data and maintaining strict confidentiality regarding their personal and financial information.

The target population comprises residential household owners and tenants residing within the MMR who are responsible for their household or community electricity bills.

The study utilizes a combination of convenience and snowball sampling. The survey was circulated among residential groups and community networks, allowing respondents to participate based on their availability and willingness, and further forward the instrument to other residents in their respective housing societies.

For accurate representation of Mumbai's unique housing landscape, the sampling strategy categorizes respondents into two distinct residential profiles:

1. Flats / Apartments in Cooperative Housing Societies (CHS).
2. Independent Houses / Bungalows / Row Houses.

The survey framework is designed to capture the following critical data points:

- a. Residential Profiling & Current Adoption Status: Identifying the respondent's housing type and determining whether their residence (or CHS common area) currently operates a rooftop solar panel system.
- b. Financial Metrics: Gathering data on the household's average monthly electricity expenses, categorized into expenditure brackets (e.g., Below ₹1,500; ₹1,500 - ₹3,000; ₹3,001 - ₹5,000; Above ₹5,000).
- c. Barrier Assessment (For Non-Adopters): Pinpointing the primary bottlenecks preventing adoption. For CHS residents, this includes high upfront costs, lack of roof access, lack of awareness, and society/landlord objections. For independent homeowners, it explores space constraints, structural issues, and technological hesitancy.
- d. Policy Awareness: Measuring the penetration of government communication by asking respondents about their specific awareness of the "PM Surya Ghar: Muft Bijli Yojana" and its maximum subsidy provision of up to ₹78,000.
- e. Consumer Readiness & Willingness: Evaluating future adoption potential by asking independent homeowners if they would install a system given a proven 3-4 year payback period, and asking CHS residents if they would be willing to sign a petition urging their society committee to adopt shared solar for common areas.

To contextualize the primary field data, secondary data was extensively reviewed. This includes policy documents from the Ministry of New and Renewable Energy (MNRE),

operational guidelines from local DISCOMs (such as the BEST Undertaking), and peer-reviewed academic literature regarding the behavioral economics of renewable energy transitions in urban Indian settings.

3.3 Data Analysis Tools

The primary data collected from the field survey will be cleaned and tabulated using spreadsheet software. Descriptive statistics, including percentages and frequency distributions, will be utilized to analyze demographic profiles, gauge awareness levels of government initiatives, and categorize the primary behavioral and structural barriers to adoption. To evaluate the financial viability of transitioning to solar energy, the study will incorporate the Time Value of Money (TVM). The average monthly electricity bills will be cross-referenced with the current Central Financial Assistance (CFA) subsidy slabs to construct a Discounted Cash Flow (DCF) model. This model will apply a conservative discount rate (e.g., 8%) to represent the opportunity cost of capital, allowing for the calculation of the Discounted Payback Period and the Net Present Value (NPV) over the 25-year lifespan of a standard solar rooftop system.

Chapter 4: Field Work Observations and Analysis

This chapter presents the empirical findings derived from the primary data collection conducted within the Mumbai Metropolitan Region (MMR). Based on a survey administered to residential consumers, the analysis shows current rooftop solar adoption rates, household electricity expenditures, and the behavioral and financial barriers hindering wider implementation. The sample encompasses a diverse mix of living arrangements, including residents of Flats in Cooperative Housing Societies (CHS) and Independent Houses/Bungalows.

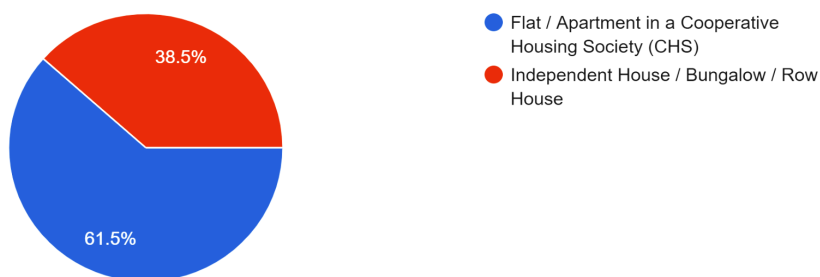
4.1 Demographic and Housing Profile and Adoption Status

A total of 13 valid responses were recorded, with all respondents providing explicit consent to participate in the academic research. The respondents were categorized based on their residence type. The sample provides a representative split between multi-tenant buildings and individual homes:

- a. Flat / Apartment in a Cooperative Housing Society (CHS): 8 respondents (61.5%)
- b. Independent House / Bungalow / Row House: 5 respondents (38.5%)

This distribution reflects the heavy prevalence of Cooperative Housing Societies in the MMR, where collective decision-making plays a crucial role in infrastructure upgrades.

Figure 4.1: *Distribution of Respondents by Residence Type*



The survey identified the current status of rooftop solar adoption among these two groups:

- a. Overall Adoption: Out of the 13 surveyed households, 5 respondents (approx. 38.5%) reported that they currently benefit from a rooftop solar system.
- b. CHS Adoption: Among the 8 CHS residents, 3 respondents (37.5%) indicated that their building society currently has a solar panel system installed to power common areas (such as lifts, water pumps, and staircase lights).
- c. Independent House Adoption: Among the 5 independent homeowners, 2 respondents (40%) have already installed a rooftop solar panel system at their residence.

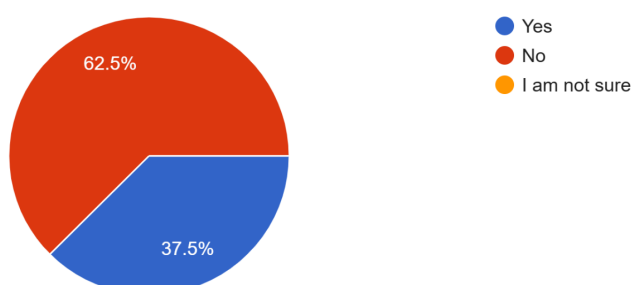


Figure 4.2.1: *Shared Solar Adoption Status in Cooperative Housing Societies*

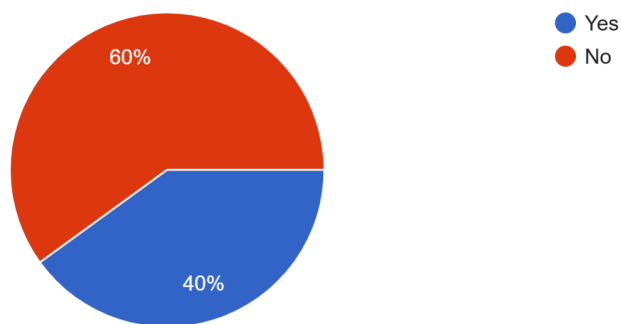


Figure 4.2.2: Rooftop Solar Adoption Status Among Independent Homeowners

4.2 Analysis of Barriers for Non-Adopters

Out of the 13 surveyed respondents, 8 households (comprising 5 CHS residents and 3 independent homeowners) currently do not have a rooftop solar system installed. Their responses reveal a multifaceted web of financial, informational, structural, and social barriers.

Figure 4.3.1: Monthly Electricity bill of Non-Adopting CHS Residents

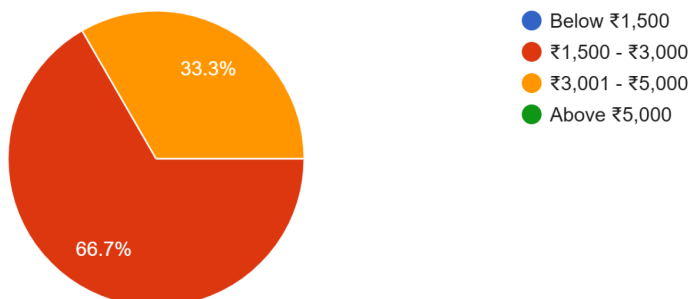
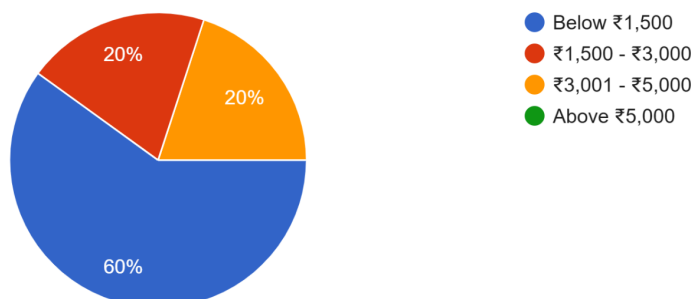


Figure 4.3.2: Monthly Electricity bill of Non-Adopting Independent Homeowners

4.2.1 The Upfront Cost

Despite the long-term savings associated with zero or reduced electricity bills, the initial capital expenditure remains the most prominent barrier. Among the non-adopters, 3 out of the 8 respondents (nearly 38%) cited "High upfront installation cost" as the primary reason for not installing solar panels. This sentiment was shared across both housing profiles, one CHS resident and two independent homeowners, indicating that even with government subsidies available the out-of-pocket investment required before the subsidy is disbursed continues to deter middle-income households (particularly those in the ₹1,500 - ₹3,000 monthly bill bracket).

4.2.2 Information and Awareness Deficits

A secondary barrier identified in the survey is a systemic information asymmetry regarding the procedural execution of the PM Surya Ghar: Muft Bijli Yojana. While the government has launched campaigns to promote the scheme, the primary data reveals a critical gap between policy awareness and actionable consumer-level knowledge. Two respondents explicitly cited informational deficits:

- a. One CHS resident pointed to a "Lack of awareness about the process and benefits."
- b. One independent homeowner highlighted a "Lack of awareness about the installation process and vendors."

This data aligns with broader industry findings that while people might be generally aware of solar energy, procedural steps remain opaque and overwhelming to the average consumer.

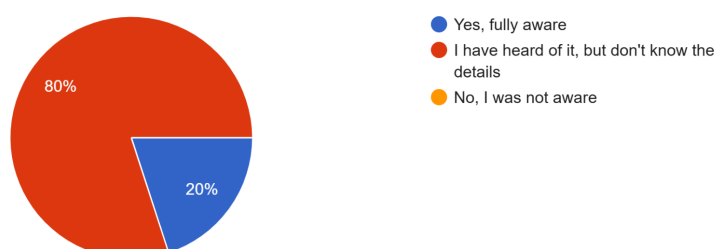


Figure 4.4.1: Awareness of the PM Surya Ghar Yojana Among Non-Adopting CHS Residents

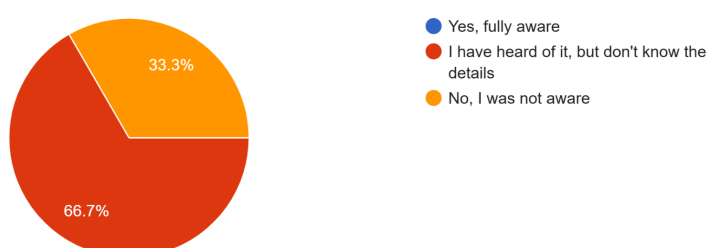


Figure 4.4.2: Awareness of the PM Surya Ghar Yojana Among Non-Adopting Independent Homeowners

80% of CHS residents and 66.7% of independent homeowners reported that they "have heard of the scheme, but don't know the details." These findings perfectly align with broader industry observations regarding India's solar transition. Information asymmetries and consumer perceptions further inhibit adoption. Ultimately, the data indicates that while

top-line policy announcements for the PM Surya Ghar Yojana successfully generate initial interest, the procedural steps remain opaque. Navigating the National Portal, identifying empaneled vendors, understanding net-metering agreements, and calculating exact Central Financial Assistance (CFA) subsidies are still perceived as overwhelming tasks for the average MMR consumer. Without step-by-step guidance, this information deficit causes potential adopters to abandon the transition process entirely.

4.2.3 Structural and Spatial Limitations

One CHS respondent cited "Lack of physical space / No private roof access" as the main reason their society has not adopted solar. In high-rise apartments, the available terrace space is often heavily occupied by water tanks, elevator machine rooms, and HVAC infrastructure, leaving minimal shadow-free areas for optimal solar generation. Furthermore, individual flat owners typically do not have private roof access to install personal systems, forcing them to rely on shared society infrastructure.

One CHS respondent identified "Society committee/landlord objections" as the primary barrier. In Cooperative Housing Societies, installing a solar system requires consensus among residents and formal approval from the managing committee. As highlighted in the broader literature regarding behavioral transitions, this requires a resident to take on the political burden of championing the project, addressing the collective risk aversion of the committee, and securing a majority vote which often stalls due to trust gaps or internal society politics.

In summary, the data clearly indicates that while financial risk remains the dominant concern, targeted interventions in Mumbai must equally address the informational gaps regarding vendor processes and the complex social dynamics inherent to Cooperative Housing Societies.

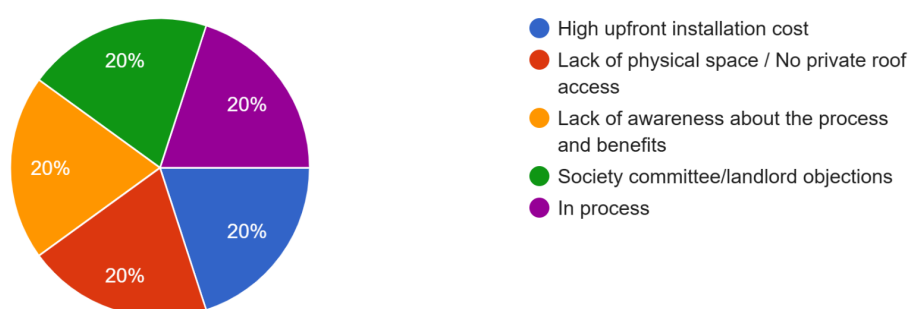


Figure 4.5.1:
Barriers to Shared Solar Adoption in CHS

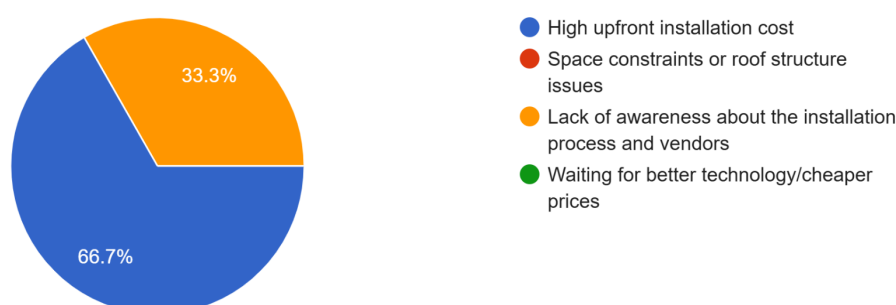


Figure 4.5.2: Barriers to Shared Solar Adoption for Independent Homeowners

HOW YOUR BILL WAS CALCULATED		
Heads	Rate/Tariff	INR(₹)
Electrical Energy (HSN Code 27160000)		
(Rs./unit)		
1. Demand/Fixed Charge		135.00
2. Wheeling Charges		763.85
3. Green Power Charges		0.00
4. Energy Charge		1498.00
5. Time of the day 00- 06 Hrs charge	0.00	0.00
6. Time of the day 06- 09 Hrs charge	0.00	0.00
7. Time of the day 09- 17 Hrs charge	0.00	0.00
8. Time of the day 17- 22 Hrs charge	0.00	0.00
9. Time of the day 22- 24 Hrs charge	0.00	0.00
10. Power Purchase Cost Adjustment (PPCA)	Refer Tariff structure	124.94
11. Government Electricity Duty	16.00%	403.49
12. Mah Govt Tax on sale of electricity	0.3594	100.63
(A) Current month's bill amount (Sum of 1-12)		3025.91
a. Others/SD interest/VDS interest		0.00
b. Delayed Payment Charge (DPC) levied		32.72
c. Digital Payment Discount		5.47Cr
d. Interest on Arrears		0.00
e. Adjustments		10.00Cr
(B) Net other charges in current bill (Sum of a to e)		17.25
Total current month charges (A+B)		3043.16
Previous month's bill amount		2636.05
Payment received up to (06-04-2026 via Online)		2630.00
Prompt/Advance payment discount		0.00
TDS Credit/Other Payment		0.00
(C) Net previous balance		6.05
Total bill amount (A+B+C)		3049.21

Figure 4.6: Sample Monthly Electricity Bill of a Non-Adopting CHS Household

4.3 Cost-Benefit Analysis

To directly address the behavioral barriers identified in the survey specifically for the non-adopting respondents who cited "High upfront installation cost" and the individuals who stated they needed to "see the exact math", this section provides a projected financial model.

Based on the survey data, the household electricity bills for non-adopters are distributed across the lower-to-middle tiers: 3 respondents pay below ₹1,500, 3 pay between ₹1,500 and ₹3,000, and 2 pay between ₹3,001 and ₹5,000. To effectively offset the consumption of the majority (those paying ₹3,000 or less), a 2 kW on-grid solar system is the most optimal and cost-effective capacity.

4.3.1 Capital Expenditure and Subsidy

The financial dynamics of residential solar adoption have shifted significantly with the introduction of the PM Surya Ghar: Muft Bijli Yojana, which aims to provide up to 300 units of free electricity monthly. The scheme alters the consumer's capital expenditure through a structured Central Financial Assistance (CFA) mechanism. For a 2 kW capacity system, consumers are eligible for a subsidy of ₹30,000 per kW, amounting to a total subsidy of ₹60,000.

In the current market, the gross cost of a high-quality 2 kW on-grid system including panels, inverters, mounting structures, and net-metering setup averages around ₹1,20,000. Applying the PM Surya Ghar subsidy brings the net out-of-pocket investment down to approximately ₹60,000. Furthermore, the government facilitates collateral-free, low-interest loans (around 7%) to further ease this initial financial burden.

4.3.2 Operational Savings, Discounted Payback Period, and NPV

A standard 2 kW system generates roughly 240 to 300 units of electricity per month. For the majority of the surveyed households in the MMR, this generation capacity is sufficient to completely offset a monthly electricity bill of up to ₹2,000 through net-metering agreements yielding annual savings (C_t) of ₹24,000.

To evaluate the true economic viability of this 2 kW solar system, a Discounted Cash Flow (DCF) analysis is conducted. A discount rate (r) of 8% is applied to represent the opportunity cost of capital equivalent to a standard long-term fixed deposit yield in India. Therefore the Discount Factor for any year (t) is calculated as:

$$\text{Discount Factor} = \frac{1}{(1+r)^t}$$

This discount factor is multiplied to the net cash flow (₹24,000) to determine the Present Value (PV) for each year which is summed sequentially to calculate the Cumulative PV.

As illustrated in the table below (Table 4.1), the Cumulative PV starts at an initial deficit of -₹60,000. By observing the highlighted row, we can clearly identify the Discounted Payback Period. The Cumulative PV flips from a deficit to a surplus (+₹1,850.33) at the end of Year 3, proving that the system mathematically breaks even and recovers its initial capital cost within a three-year timeframe.

Year (t)	Net Cash Flow	Discount Factor	Present Value	Cumulative PV
0	-₹60,000.00	100.00%	-₹60,000.00	-₹60,000.00
1	₹24,000.00	92.59%	₹22,222.22	-₹37,777.78
2	₹24,000.00	85.73%	₹20,576.13	-₹17,201.65
3	₹24,000.00	79.38%	₹19,051.97	₹1,850.33
4	₹24,000.00	73.50%	₹17,640.72	₹19,491.04
5	₹24,000.00	68.06%	₹16,334.00	₹35,825.04
...	...	...	...	...
25	₹24,000.00	14.60%	₹3,504.43	₹196,194.63
NPV:				₹196,194.63

Table 4.1: Discounted Cash Flow (DCF) Model and Break-Even Analysis

Since the system's operational lifespan is 25 years, the Cumulative PV at the 25th year represents the total Net Present Value (NPV) of the investment. This lifetime wealth generation can be directly calculated using the Present Value of an Annuity (PVA) formula:

$$NPV = -C_0 + \sum_{t=1}^n \frac{C_t}{(1+r)^t}$$

Where:

C_0 = Initial Net Investment (₹60,000)

C_t = Annual Cash Inflow / Savings (₹24,000)

r = Discount Rate (8%)

n = Lifespan of the system (25 years)

Calculating the Present Value of an Annuity (PVA) for the 25-year period:

$$NPV = -60000 + 24000 \times \left[\frac{1-(1+0.08)^{-25}}{0.08} \right]$$

$$NPV = -60000 + 24000 \times 10.6748$$

$$NPV = -60000 + 256195 = ₹1,96,195$$

The highly positive NPV of ₹1,96,195 proves that investing in a 2 kW rooftop solar system under the PM Surya Ghar scheme is profoundly economically viable. Not only does the system recover its discounted costs within three years, but it also generates nearly ₹2 Lakhs in real wealth in today's rupees over its operational life.

4.3.3 Sensitivity Analysis on the Discount Rate

To test the robustness in the Discounted Cash Flow (DCF) model, a one-way sensitivity analysis was conducted on the discount rate (r).

The baseline model utilizes an 8% discount rate to reflect the opportunity cost of capital based on standard long-term fixed deposit yields. However, macroeconomic fluctuations, inflation, or alternative investment opportunities can alter this cost. Table 4.2 illustrates how the Net Present Value (NPV) responds if the discount rate is adjusted to a more optimistic 6% (for lower market interest rates) or a more conservative 10% (for higher alternative yields or inflation).

Scenario	Discount Rate	NPV
Optimistic	6.00%	₹246,800.55
Baseline	8.00%	₹196,194.63
Conservative	10.00%	₹157,848.96

Table 4.2: Sensitivity of NPV to Variations in the Discount Rate

4.3.4 Reconciling Findings with Literature

The Cost-Benefit Analysis projects annual savings of ₹24,000 and a discounted payback period of under 3 years which presents a more optimistic financial outlook compared to the literature. For instance, Shah (2024) estimated annual savings under the same scheme to be between ₹15,000 and ₹18,000, while Sharma estimated a broader payback window of 5 to 10 years for standard rooftop installations. This variance can be a reflection of three distinct economic variables specific to this study's parameters:

1. The savings in solar models are dictated by the retail cost of the grid electricity being replaced. While the survey data indicates that most non-adopting households fall into the "below ₹1,500" or "₹1,500 to ₹3,000" billing brackets, the electricity tariffs in Mumbai (inclusive of high base rates, wheeling charges, and state duties) remain notably steeper than the national average. Therefore, Shah's (2024) lower savings estimate of ₹15,000 to ₹18,000 likely reflects national averages or regions where these lower-to-middle consumption slabs are heavily subsidized by local distribution companies.
2. The DCF model assumes the installation of a 2 kW system because its generation capacity perfectly offsets a ₹2,000 monthly bill for these surveyed households. By achieving a near 100% offset of the bill without generating massive excess surplus (which typically earns a much lower feed-in tariff from the DISCOM), the model maximizes the financial return, yielding exactly ₹24,000 in annual savings. Broader literature estimating longer payback periods (Sharma) might account for sub-optimal system sizing, where households install larger capacities that suffer from lower returns.
3. The three year payback period leverages the architecture of the scheme which aggressively subsidizes a 2 kW system. Sharma's 5-to-10-year payback window likely accounts for larger capacity systems (where the subsidy strictly caps out at ₹78,000, resulting in diminishing marginal subsidy returns) or utilizes pre-2024 capital expenditure baselines before panel prices dropped globally.
4. By modeling a 2 kW system designed for the average consumption profile of our non-adopting survey respondents (bills of ₹2,000 to ₹2,500), the model minimizes excess generation. Excess generation fed back into the grid often earns a lower tariff from DISCOMs than the retail rate, which can artificially drag out the payback period in improperly sized systems.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

The transition toward decentralized renewable energy is a critical pillar of India's sustainable future, although achieving widespread residential rooftop solar (RTS) adoption in a dense urban landscape like Mumbai remains a complex challenge. This field project evaluated the true cost-benefit dynamics and the behavioral, structural, and administrative barriers influencing solar adoption among MMR households, specifically within the context of the ambitious PM Surya Ghar: Muft Bijli Yojana. Through the primary survey data and secondary literature, several key conclusions emerge:

1. The financials for residential rooftop solar is highly favorable. With the Central Financial Assistance (CFA) provided under the PM Surya Ghar scheme offering up to ₹60,000 for a 2 kW system and strictly capped at ₹78,000 for systems 3 kW and above, the net out-of-pocket capital expenditure for consumers is drastically reduced. As demonstrated in the Cost-Benefit Analysis, a 2 kW system generating annual savings of ₹24,000 achieves a discounted payback period of under 3 years. It also yields a highly positive Net Present Value of ₹1,96,195 by securing free electricity for the remainder of the system's 25-year lifespan.
2. Despite these long-term savings and available subsidies, the initial capital requirement continues to be perceived as the most significant obstacle. The primary data reflects that nearly 38% of non-adopting respondents cited "High upfront installation cost" as a barrier. This indicates that the mere existence of subsidies is not enough without transparent financial communication.
3. A majority of the population resides in Cooperative Housing Societies (CHS), where high-rise buildings limit private roof access and available shadow-free space. Consequently, RTS adoption in Mumbai is highly dependent on collective decision-making. The survey and literature underscore that social dynamics, committee objections, and an unwillingness to take responsibility for initiating community-level changes stall adoption.
4. Consumers suffer from a lack of awareness regarding the end-to-end installation process, vendor selection, and subsidy claiming. Furthermore, a significant trust deficit exists toward institutional claims, vendors, and the operational reliability of the systems. Overcoming this requires building trust through local community relationships, and transparent performance data.
5. While RTS minimizes transmission losses and provides grid resilience, DISCOMs often perceive residential solar as a threat to their revenue streams due to lost energy sales and increased cross-subsidy burdens. Additionally, the operational complexities of net-metering, billing, and integrating variable solar generation place an administrative strain.

Ultimately, while the PM Surya Ghar Yojana has provided a financial architecture to make solar energy accessible and affordable, achieving scale in the MMR requires going beyond financial incentives. Widespread adoption relies on bridging the gap between national policy and consumer psychology. It necessitates addressing the structural hurdles of apartment living, simplifying the administrative pipeline for CHS approvals, empowering DISCOMs to act as active facilitators, and deploying targeted awareness campaigns to translate financial math for the everyday household.

5.2 Recommendations

5.2.1 For Policy Makers

1. Policymakers must formalize and promote group net metering and virtual net metering models. This allows generation from a single shared system to be apportioned across multiple consumer accounts, and enables residents to subscribe to off-site solar installations.
2. Establishing transparent regulations alongside enforceable service standards for utilities will provide the predictability necessary for households to commit to long-term solar investments.
3. While the PM Surya Ghar scheme's subsidies are generous, ensuring that disbursements are simple, timely, and predictable is critical to reducing consumer hesitation and enhancing vendor liquidity.

5.2.2 For Distribution Companies (DISCOMs)

1. To mitigate the perceived threat of revenue loss, DISCOMs like BEST and Mahavitaran should adopt models such as Utility-Led Community Solar, On-Bill Financing, or Solar Partners that can help DISCOMs capture benefits from the solar market.
2. DISCOMs can act as facilitators by aggregating demand among interested residential societies like the BRPL Solar City Initiative. DISCOMs can invite bids from empaneled vendors, reducing the project cost for consumers and lowering the customer acquisition risk for developers.
3. DISCOMs must invest in training their personnel to handle the administrative load of processing applications, inspections, and net-metering billing. Furthermore, conducting power system simulation studies on distribution transformers will help prepare the local grid for higher solar penetration without compromising safety or power quality.

5.2.3 For Solar Vendors

1. The primary data identifies that non-adopters suffer from decision paralysis due to financial uncertainty. Vendors should provide financial pathways, such as clear payback period charts, lifetime ROI estimates, and easy-to-understand breakdowns of the PM Surya Ghar subsidies.
2. Consumers trust local relationships more than institutional claims. Vendors should aim to establish long-term service relationships and provide transparent lifecycle performance data to build credibility.
3. Utilizing neighborhood scoreboards or sharing local case studies of successful installations can transform rooftop solar from an abstract concept into a tangible, socially validated reality, encouraging more housing societies to participate.

5.2.4 For Cooperative Housing Societies

1. Housing societies should focus first on installing shared systems to power common area amenities (lifts, water pumps, staircase lighting). This utilizes the specific ₹18,000 per kW subsidy for GHS/RWA common facilities, offering a highly visible reduction in monthly society maintenance charges and serving as a proof-of-concept for skeptical residents.
2. Committees should facilitate resident-led meetings or informal discussions to address doubts, explain warranties, and collectively evaluate vendor proposals before calling for a vote.

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Appendix

Appendix A: Field Survey Questionnaire

Section A: Consent and Housing Profile

1. Consent to Participate

Do you explicitly consent to participate in this academic research study? (Your responses will remain anonymous and will be used solely for academic data analysis).

☐ Yes (Proceed to Q2)

☐ No (Terminates survey)

2. What type of residence do you currently live in?

☐ Flat / Apartment in a Cooperative Housing Society (CHS) (Proceed to Section B)

☐ Independent House / Bungalow / Row House (Skip to Section C)

Section B: For Flat / Apartment Residents (CHS)

3. Does your building society currently have a rooftop solar panel system installed to power common areas (lifts, water pumps, staircase lights)?

☐ Yes (End of survey)

☐ No (Proceed to Q4)

☐ I am not sure (Proceed to Q4)

4. What is your household's average monthly electricity bill?

☐ Below ₹1,500

☐ ₹1,500 - ₹3,000

☐ ₹3,001 - ₹5,000

☐ Above ₹5,000

5. What do you think is the primary reason your housing society has NOT installed solar panels for common areas?

☐ High upfront installation cost

☐ Lack of physical space / No private roof access

☐ Lack of awareness about the process and benefits

☐ Society committee / landlord objections

☐ In process

6. Are you aware of the government's "PM Surya Ghar: Muft Bijli Yojana" which provides subsidies of up to ₹78,000 for residential solar installations?

☐ Yes, fully aware

☐ I have heard of it, but don't know the details

☐ No, I was not aware

(End of Survey for CHS Residents)

Section C: For Independent Houses / Bungalows

7. Do you currently have a rooftop solar panel system installed at your residence?

☐ Yes (End of survey)

☐ No (Proceed to Q8)

8. What is your average monthly electricity bill?

- ☐ Below ₹1,500
- ☐ ₹1,500 - ₹3,000
- ☐ ₹3,001 - ₹5,000
- ☐ Above ₹5,000

9. What is the primary reason you have NOT installed solar panels?

- ☐ High upfront installation cost
- ☐ Space constraints or roof structure issues
- ☐ Lack of awareness about the installation process and vendors
- ☐ Waiting for better technology / cheaper prices

10. Are you aware of the government's "PM Surya Ghar: Muft Bijli Yojana" which provides subsidies of up to ₹78,000 for residential solar installations?

- ☐ Yes, fully aware
 - ☐ I have heard of it, but don't know the details
 - ☐ No, I was not aware
- (End of Survey for Independent House Residents)

Appendix B: 25-Year Discounted Cash Flow (DCF) Model and Sensitivity Analysis

Assumptions	
Discount Rate	8.00%
Initial Investment	-₹60,000.00
Annual Savings	24,000.00
System Lifespan	25

Year (t)	Net Cash Flow	Discount Factor	Present Value (PV)	Cumulative PV
0	-₹60,000.00	100.00%	-₹60,000.00	-₹60,000.00
1	₹24,000.00	92.59%	₹22,222.22	-₹37,777.78
2	₹24,000.00	85.73%	₹20,576.13	-₹17,201.65
3	₹24,000.00	79.38%	₹19,051.97	₹1,850.33
4	₹24,000.00	73.50%	₹17,640.72	₹19,491.04
5	₹24,000.00	68.06%	₹16,334.00	₹35,825.04
6	₹24,000.00	63.02%	₹15,124.07	₹50,949.11
7	₹24,000.00	58.35%	₹14,003.77	₹64,952.88
8	₹24,000.00	54.03%	₹12,966.45	₹77,919.33
9	₹24,000.00	50.02%	₹12,005.98	₹89,925.31
10	₹24,000.00	46.32%	₹11,116.64	₹101,041.95
11	₹24,000.00	42.89%	₹10,293.19	₹111,335.14
12	₹24,000.00	39.71%	₹9,530.73	₹120,865.87
13	₹24,000.00	36.77%	₹8,824.75	₹129,690.62
14	₹24,000.00	34.05%	₹8,171.06	₹137,861.69
15	₹24,000.00	31.52%	₹7,565.80	₹145,427.49
16	₹24,000.00	29.19%	₹7,005.37	₹152,432.86
17	₹24,000.00	27.03%	₹6,486.45	₹158,919.31
18	₹24,000.00	25.02%	₹6,005.98	₹164,925.29
19	₹24,000.00	23.17%	₹5,561.09	₹170,486.38
20	₹24,000.00	21.45%	₹5,149.16	₹175,635.54
21	₹24,000.00	19.87%	₹4,767.74	₹180,403.28
22	₹24,000.00	18.39%	₹4,414.57	₹184,817.85
23	₹24,000.00	17.03%	₹4,087.57	₹188,905.41
24	₹24,000.00	15.77%	₹3,784.78	₹192,690.20
25	₹24,000.00	14.60%	₹3,504.43	₹196,194.63
NPV:				₹196,194.63

Scenario	Discount Rate	NPV
Optimistic	6.00%	₹246,800.55
Baseline	8.00%	₹196,194.63
Conservative	10.00%	₹157,848.96